

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

E. Constructive Xor

time limit per test: 1 second
memory limit per test: 256 megabytes

You are given a sequence of n integers $a_1, a_2, \dots, a_n$.

Your task is to answer q queries:

- Choose several numbers from $a_1, a_2, \dots, a_n$ such that their bitwise XOR is x . If their are multiple solutions, print any of them.

It's guaranteed that at least one solution exists for each query.

Input

The first line of input contains a single integer n ($1 \leq n \leq 500$).

The second line contains n integers $a_1, \dots, a_n$ ($0 \leq a_i < 2^{60}$).

Then, the third line contains a single integer q ($1 \leq q \leq 1000$).

For the next q lines, each line contains a single integer x ($0 \leq x < 2^{60}$) denoting a query.

Output

For each query, print a line of n zeros and/or ones. The i -th character should be '1' if the solution contains a_i , or should be '0' otherwise.

Example

input	Copy
6 2 15 12 0 5 4 5 10 8 3 6 7	
output	Copy
010010 110010 011000 011010 100010	

Introductory Problems: XOR Basis

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
327038515	Jul/02/2025 09:59	Accepted
327037676	Jul/02/2025 09:52	Accepted
327037611	Jul/02/2025 09:52	Accepted
327036916	Jul/02/2025 09:45	Wrong answer on test 1

